EXACT TORSION GROUPS REALIZED ON GEOMETRICALLY SPLIT GENUS-2 JACOBIANS OVER $\mathbb{Q}$

Groups are written by invariant factors: $[n_1, \ldots, n_r]$ denotes $\mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_r\mathbb{Z}$ with $n_1 \mid \cdots \mid n_r$. Each row records one genus-2 curve $C/\mathbb{Q}$ whose Jacobian is geometrically *split* — isogenous over $\overline{\mathbb{Q}}$ to a product of two elliptic curves, i.e. not geometrically simple — and whose *full* rational torsion group is *exactly* the displayed group, $\mathrm{Jac}(C)(\mathbb{Q})_{\mathrm{tors}} \cong G$ (the first row is the trivial group, written $[\,]$). This is the split companion of `torsion_realizations.tex`; the two tables partition the known realizations by geometric (non)simplicity. We display, for each group, the equation of the curve of smallest conductor in the queried extended-database snapshot (2026-08-12) with Jac geometrically split; rows whose group does not occur in the database show a construction from the literature or from the present paper. Curves of small conductor and discriminant lie in the production LMFDB; the remaining database curves cite [2]. For convenience each database equation is hyperlinked to its LMFDB home page: production curves by their permanent label, and extended-database curves via an alpha-site (`https://alpha.lmfdb.org`) equation-lookup URL, since alpha labels are not permanent (the lookup matches any isomorphic model of the curve). The equations are the intended stable identifiers.

Source(s) column. It gives the earliest construction of the torsion group on a split genus-2 Jacobian over $\mathbb{Q}$, together with the source of an exact-torsion witness when these differ; "new" means the exact split realization first appears in the present paper (witness equations, split certificates, and elliptic factors are recorded in `product/data/new_split_witnesses.txt` and the verification logs cited there). A dagger (†) marks a source that exhibits the torsion — typically a positive-dimensional family with G *contained* in the torsion (Howe–Leprévost–Poonen, Theorem 1) — without an exact-torsion witness. Six rows are marked "new": $[45]$, whose exact witness comes from an earlier stage of this project (verified in `results/claude_ov_lane8_verify.log`), and $[8, 8]$, $[6, 12]$, $[2, 2, 24]$, $[2, 2, 4, 8]$, $[2, 2, 4, 4]$, constructed by this paper's gluing search — $[2, 2, 24]$ and $[2, 2, 4, 8]$ by instantiating the square conditions of [3, §3.7], and $[2, 2, 4, 4]$ by gluing the elliptic curves `210.c5` and `2310.o4` (its Jacobian has a Richelot isogeny to their product). All six are archived in `product/data/new_split_witnesses.txt`. Five of the six ($[45]$, $[8, 8]$, $[6, 12]$, $[2, 2, 24]$, $[2, 2, 4, 8]$) lie in HLP's Table 1, where they were previously containment-only, so every group of that table now has an exact witness.

The $[8, 8]$ witness is one member of an apparently infinite exact family: on the $X_1(8) \times X_1(8)$ gluing chart the parameter locus $t = a^2/(a^2 + b^2)$ satisfies the gluing conditions identically, and all 59 distinct instances found to parameter height 250 have exact torsion $[8, 8]$ (HLP's family gives the containment; the observed exactness throughout the family is new — see `product/logs/lane1212_H250.log`).

Two groups occur only for $\mathbb{Q}$-simple, geometrically split Jacobians: $[19]$ (the modular Jacobian $J_1(13)$, [6]) and $[25]$; every other row is split over $\mathbb{Q}$. The largest order occurring is $128 = \#[2, 2, 4, 8]$ (the maximum of HLP's Table 1), and the largest cyclic order is 70 [4]. The $[36]$ and $[48]$ rows are the curves of Platonov–Petrunin [9]: the displayed minimal-conductor database witnesses coincide with their curves f_{36} and $f_{48,1}$ (and the second curve of [9] is the database curve `5292.c2`), with geometric non-simplicity proven in [8, §7].

Group	Curve	Source(s)
$[\,]$	$y^2 + y = -x^6 + 3x^5 - 50x^4 + 95x^3 - 14x^2 - 33x - 6$	[1]
$[2]$	$y^2 + (x^3 + x)y = -x^6 + 15x^4 - 75x^2 - 56$	[1]
$[3]$	$y^2 + (x^3 + x + 1)y = -54x^6 - 5x^5 + 302x^4 + 92x^3 - 479x^2 - 318x - 54$	[2]
$[4]$	$y^2 + (x^3 + x^2)y = -4x^6 + 76x^5 - 304x^4 - 744x^3 - 4239x^2 - 4824x - 7536$	[2]
$[5]$	$y^2 = -8x^6 - 36x^5 + 14x^4 + 17x^3 - 334x^2 + 348x - 344$	[2]

Date: August 2026.

Group	Curve	Source(s)
[6]	$y^2 + (x^2 + x)y = -5x^6 - 3x^5 - 12x^4 + 37x^3 + 3x^2 + 42x - 92$	[2]
[7]	$y^2 + (x^3 + 1)y = -3x^6 - 4x^5 + 2x^4 + 7x^3 + 2x^2 - 4x - 3$	[2]
[8]	$y^2 + (x^3 + x)y = -4x^4 + 7x^2 - 15$	[2]
[9]	$y^2 + (x^3 + x)y = -x^5 - 2x^4 + x^3 - 5x^2 + x - 3$	[2]
[10]	$y^2 + x^2y = -11x^5 - 13x^4 + 7x^3 + 10x^2 - x - 2$	[1]
[12]	$y^2 + (x^3 + 1)y = x^4 - x^3 + x^2$	[1]
[14]	$y^2 + (x^3 + x^2)y = -x^5 + x^4 - 3x^3 + x^2 - 4x$	[2]
[15]	$y^2 + x^3y = x^4 + 2x^2 + 1$	[1]
[16]	$y^2 + (x^3 + x)y = 2x^6 - 26x^4 + 97x^2 - 125$	[2]
[18]	$y^2 + (x^3 + x^2)y = -x^5 - x^4 + 2x^3 - 7x^2 + 5x - 6$	[2]
[19]	$y^2 + (x^3 + x^2 + 1)y = x^2 + x$	[6, 7]
[20]	$y^2 + (x^3 + x)y = x^5 - 2x^4 + 3x^3 - 7x - 2$	[3]†, [2]
[21]	$y^2 + (x^3 + x + 1)y = x^5 + 2x^4 + 2x^3 + x^2$	[7]†, [5]
[24]	$y^2 + (x^3 + x)y = 8x^4 + 97x^2 + 375$	[5]
[25]	$y^2 + (x^3 + x^2)y = 2x^6 + 4x^5 + 14x^4 + 9x^3 + 13x^2 + 8x + 1$	[5]†, [10]
[27]	$y^2 + (x + 1)y = x^6 + 9x^4 + 17x^3 + 20x^2 + 4x$	[5]
[28]	$y^2 + (x^2 + x)y = x^6 - 3x^5 + 5x^4 - x^3 - 4x^2 + 10x + 4$	[4]
[30]	$y^2 + (x^3 + 1)y = 2x^6 + 6x^5 + 10x^4 + 11x^3 + 10x^2 + 6x + 2$	[3]†, [2]
[35]	$y^2 + (x^2 + x)y = 10x^6 + 30x^5 + 116x^4 + 182x^3 + 166x^2 + 80x + 160$	[3]†, [2]
[36]	$y^2 + (x^2 + x)y = x^6 - x^5 + 5x^4 - 5x^3 - x^2 - 9x + 9$	[9, 8]
[40]	$y^2 + (x^3 + x)y = 2x^4 + 12x^2 + 66$	[3]†, [2]
[45]	$y^2 = 13981x^6 + 29240200x^4 + 49996210000x^2 + 168300000000$	[3]†, new
[48]	$y^2 + (x^3 + x)y = -4x^4 + 9x^2 + 3$	[9, 8], [4]
[60]	$y^2 + (x^2 + x)y = x^6 - 7x^5 + 25x^4 - 24x^3 + 25x^2 - 7x + 1$	[3]†, [2]
[63]	$y^2 = 897x^6 - 197570x^4 + 79136353x^2 - 146398496$	[3]
[70]	$y^2 + (2x^3 - 3x^2 - 41x + 110)y = x^3 - 51x^2 + 425x + 179$	[4]
[2, 2]	$y^2 + (x^3 + x^2 + x + 1)y = -60x^6 + 59x^4 - x^3 + 59x^2 - x - 60$	[2]
[2, 4]	$y^2 + (x^2 + 1)y = 3x^5 + x^4 + 11x^3 - 6x^2 + 6x - 16$	[2]
[2, 6]	$y^2 + (x^2 + x)y = -2x^5 + x^4 - 3x^3 - 14x^2 + 13x - 23$	[2]
[2, 8]	$y^2 + (x^2 + 1)y = -175x^6 + 60x^5 - 305x^4 + 447x^3 - 279x^2 + 465x - 214$	[2]
[2, 10]	$y^2 + (x^3 + x^2 + x + 1)y = -x^5 - x^4 - x^3 - x^2$	[7]
[2, 12]	$y^2 + (x^3 + x^2 + x)y = 2x^5 + 4x^4 - 10x^3 - 8x^2 + 15x - 5$	[3]†, [1]
[2, 14]	$y^2 + (x^2 + x)y = 9x^6 - 3x^5 - 9x^4 - 5x^3 - 24x^2 - 6x + 10$	[2]
[2, 16]	$y^2 + (x^3 + x^2 + x + 1)y = 2x^6 + 7x^5 + 16x^4 + 18x^3 + 16x^2 + 7x + 2$	[2]
[2, 18]	$y^2 + (x^2 + x)y = x^6 - 3x^5 - 3x^4 - 5x^3 + 12x^2 + 66x + 36$	[2]
[2, 20]	$y^2 + (x^2 + x)y = x^6 - 3x^5 + 15x^4 - 20x^3 + 42x^2 - 18x$	[2]
[2, 24]	$y^2 + (x^3 + x)y = -x^4 - 6x^2 + 15$	[3]†, [2]
[2, 30]	$y^2 + (x^2 + x)y = x^6 - 5x^5 - 13x^4 + 23x^3 + 18x^2 - 36x + 12$	[2]
[2, 48]	$y^2 + (x^2 + x)y = x^6 - 13x^5 + 19x^4 + 111x^3 + 19x^2 - 13x + 1$	[2]
[3, 3]	$y^2 = -2x^6 + 6x^5 - 8x^4 + 6x^3 - 8x^2 + 6x - 2$	[1]
[3, 6]	$y^2 + (x^2 + x)y = 6x^6 + 8x^5 + 21x^4 - 13x^3 + 56x^2 - 27x + 27$	[1]
[3, 9]	$y^2 + (x^2 + x + 1)y = x^6 + 3x^5 + 5x^4 + 5x^3 - 2x$	[5]†, [1]
[3, 12]	$y^2 + (x^2 + x + 1)y = x^6 + 3x^5 - 3x^4 - 11x^3 + 6x^2 + 12x - 8$	[3]†, [2]

Group	Curve	Source(s)
[3, 24]	$y^2 + (x^2 + x)y = x^6 - 3x^5 + 5x^4 - 6x^3 - 4x^2 + 6x + 4$	[2]
[4, 4]	$y^2 + x^2y = 7x^6 - 2x^4 - 7x^2 + 448$	[2]
[4, 8]	$y^2 + (x^3 + x)y = -5x^3 - 5x^2 - 15x$	[2]
[4, 12]	$y^2 + (x^2 + x)y = 9x^6 - 27x^5 + 53x^4 - 62x^3 + 53x^2 - 27x + 9$	[2]
[4, 16]	$y^2 + (x^3 + x)y = -4x^4 - 24x^2 + 252$	[2]
[5, 5]	$y^2 + (x^3 + x^2 + x + 1)y = x^5 + 2x^4 + 5x^3 + 2x^2 + 4x - 2$	[3], [2]
[5, 10]	$y^2 + (x^2 + x)y = 9x^6 - 3x^5 + 15x^4 - 33x^3 + 11x^2 + 135x - 135$	[3]†, [2]
[6, 6]	$y^2 + (x^2 + x)y = x^6 + 3x^5 + 6x^4 + 7x^3 + 6x^2 + 3x + 1$	[3]†, [1]
[6, 12]	$y^2 = 132x^6 + 396x^5 - 6347x^4 - 13354x^3 + 75207x^2 + 81950x + 88825$	[3]†, new
[7, 7]	$y^2 = x^6 + 3025x^4 + 3232987x^2 + 869675859$	[3]
[8, 8]	$y^2 = 836x^6 + 88596x^5 + 88597x^4 + 1800118x^3 - 4045487x^2 - 4535664x + 84285504$	[3]†, new
[2, 2, 2]	$y^2 + xy = 16x^5 + 129x^4 + 268x^3 + 32x^2 + x$	[2]
[2, 2, 4]	$y^2 + (x^3 + x)y = -2x^6 + 3x^5 + 18x^3 + 7x^2 - 15x$	[2]
[2, 2, 6]	$y^2 + (x^3 + x)y = 2x^5 + 3x^4 - 3x^3 - 4x^2 + x$	[1]
[2, 2, 8]	$y^2 + (x + 1)y = 6x^5 - 4x^4 - 5x^3 + x^2 + x$	[1]
[2, 2, 10]	$y^2 + (x^2 + x)y = 9x^6 + 3x^5 - 45x^4 - 2x^3 + 66x^2 - 12x - 20$	[2]
[2, 2, 12]	$y^2 + (x^2 + x)y = -45x^5 - 55x^4 + 19x^3 + 19x^2 - 9x + 1$	[2]
[2, 2, 16]	$y^2 + (x^2 + x)y = -11x^6 - 33x^5 + 3x^4 + 61x^3 + 3x^2 - 33x + 9$	[2]
[2, 2, 24]	$y^2 = x(52316x - 156025)(2500x + 3969)(32400x^2 - 34360x + 255881)$	[3]†, new
[2, 4, 4]	$y^2 + xy = -240x^5 + 161x^4 + 40x^2 + 15x$	[2]
[2, 4, 8]	$y^2 + (x^2 + x)y = 140x^5 + 3630x^4 + 797x^3 - 10562x^2 - 1530x + 7524$	[3]†, [2]
[2, 6, 6]	$y^2 + (x^2 + x)y = 25x^6 - 5x^5 - 40x^4 + 71x^3 + 9x^2 - 54x + 18$	[3]†, [2]
[2, 2, 2, 2]	$y^2 + (x + 1)y = -4x^5 + 25x^4 - 30x^3 - 44x^2 + 41x + 26$	[2]
[2, 2, 2, 4]	$y^2 + (x^2 + x)y = -3x^5 + 16x^4 + 34x^3 - 6x^2 - 21x$	[2]
[2, 2, 2, 6]	$y^2 + (x^2 + x)y = -30x^5 - 111x^4 - 72x^3 + 39x^2 + 30x$	[2]
[2, 2, 2, 8]	$y^2 + (x^2 + x)y = x^6 + 3x^5 - 9x^4 - 23x^3 + 30x^2 + 42x - 45$	[2]
[2, 2, 4, 4]	$y^2 + (x^2 + x)y = 60x^5 + 1000x^4 - 671x^3 - 5657x^2 + 867x + 4913$	new
[2, 2, 4, 8]	$y^2 = x(x + 336100^2)(x + 835200^2)(x + 841500^2)(x + 877221^2)$	[3]†, new

References

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